

VEGROW INTERVIEW EXPERIENCE

(INTERNSHIP)

Vegrow, India's leading B2B fruits marketplace was established in 2020 by IIT alumni. Vegrow has established a strong presence in the fruits segment in India, which is a large, unorganized market that is poised for tech-led disruption due to its fragmented and localized operational structure and is characterized by the involvement of multiple intermediaries facilitating transactions.

The company maximizes farmers' income, by accurately grading produce and efficiently matching it with the most suitable demand channel. This creates a virtuous cycle, attracting more farmers through competitive pricing as well as more buyers, who seek consistency in both price and quality.

ROUND 1:

Online Hackerrank Test:

It consists of three coding questions and a SQL query to be written.

- **0/1 Knapsack Problem** - The objective was to maximize the profit from selecting artifacts for a caravan with a strict weight limit. Given the profit values and weights of the artifacts, I needed to determine the maximum profit achievable without exceeding the weight capacity. [60 points]
- **Best Time to Buy and Sell Stock** - The task was to determine the maximum profit that can be achieved by buying a crop on one day and selling it on a later day, given an array of prices representing the crop prices over several days. [20 points]
- **Candy Distribution problem** - To calculate the minimum number of rewards (or candies) to distribute among fields based on their productivity ratings, ensuring that higher-rated fields receive more rewards than their adjacent lower-rated fields. [40 points]
- **SQL** - SQL query problem where I needed to retrieve the names of students who scored higher than 75 marks. The results were ordered by the last three characters of their names, and if there were ties, they were further sorted by the student ID. [10 points]

From the test 10 students were shortlisted and eligible for the Interview Conducted at CUIC and there are 2 Technical interviews.

ROUND 2:

TECHNICAL INTERVIEW - 1: (1 hour)

1. Tell me about yourself.
2. Explain your project.
3. Tech stacks used.
4. Give me an overview of your database that you used in your project and explain me with an ER diagram.
5. He asked me how would you modify your table. He asked me to normalize it.
6. Write a query to find the privileges offered for a particular user (with reference my table).
7. Sort an array containing the numbers 1, 2 and 3 in linear time complexity.
8. He modified the question that the number in the array can be of any range restricting the time complexity to be $O(n)$. I replied that it is not feasible. Then he restricted the range to the numbers [1, 10000]. I suggested the solution using hash array. He asked me what if the array contained so many duplicates. I suggested using Sorted Hashmap. (map in C++, TreeMap in java).
9. You are left in desert with a bag containing 2 red and 2 blue pills. You must eat exactly one blue and one red to survive. How will you survive. [Hint: Think of breaking the pills]
10. Tell me about Docker.

From this round 5 students got shortlisted for the next round.

TECHNICAL INTERVIEW-2: (20mins)

1. Tell me about yourself.
2. Tell me about project.
3. Explains tables.
4. A simple SQL join query.
5. Integer to Roman conversion
6. Factorial of a number.
7. Majority element in const space complexity [Use Moore's voting algorithm].
8. Minimum number of insertions to make a string palindrome. [Note: insertion can be done only at the beginning of the string.]

Finally 3 students got selected for the internship.

MY BEST WISHES FOR YOU

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